INDIA SPACE LAB

Project Report

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PART - A

Q1. Explain Finite Element Method (FEM). Discuss what FEM is, why it is used, and its applications in aerospace engineering.

The **Finite Element Method (FEM)** is a numerical analysis technique used to solve complex engineering problems involving structures, heat transfer, vibration, and other physical phenomena. Instead of solving the entire structure as one large system, FEM divides it into many small, simple parts called **finite elements**.
These elements are connected at **nodes**, and mathematical equations are solved for each element to predict the behavior of the complete structure.

Why FEM is Used

- To analyze complex geometries that cannot be solved using analytical methods.
- To predict stress, strain, displacement, temperature, and deformation.
- To reduce the need for expensive physical prototypes.
- To improve product safety, reliability, and performance before manufacturing.

Applications in Aerospace Engineering

- Structural analysis of aircraft wings, fuselage, and landing gear.
- Stress and fatigue analysis of engine components.
- Thermal analysis of spacecraft during atmospheric re-entry.
- Vibration and modal analysis of aircraft structures.
- Composite material analysis for lightweight aerospace components.

Q2. What is Von Mises Stress and why is it important? Explain yielding and structural failure concepts.

Von Mises Stress is an equivalent stress used to determine whether a ductile material will begin to yield under complex loading conditions. It combines stresses acting in different directions into a single value that can be compared with the material's yield strength.

Importance of Von Mises Stress

- Predicts the onset of yielding in ductile materials.
- Widely used in FEM software for structural design.
- Helps engineers determine whether a component is safe under applied loads.
- Supports optimization by reducing unnecessary material while maintaining safety.

Yielding

Yielding occurs when the stress in a material exceeds its **yield strength**. At this point, the material undergoes **permanent (plastic) deformation**, meaning it will not return to its original shape after the load is removed.

Structural Failure

Structural failure occurs when a component can no longer safely carry the applied load. Failure may happen due to:

- Excessive yielding
- Fracture or cracking
- Buckling under compressive loads
- Fatigue caused by repeated loading

If the **Von Mises stress exceeds the material's yield strength**, the component is likely to yield and may eventually fail if the loading continues.

Q3. Explain the role of meshing in FEM simulations. Discuss accuracy, computational cost, and mesh dependency.

Meshing is the process of dividing a model into many small finite elements before performing an FEM analysis. The quality and size of the mesh directly affect the accuracy and efficiency of the simulation.

Accuracy

- A **finer mesh** contains smaller elements and generally produces more accurate results.
- Fine meshes better capture stress concentrations and complex geometries.
- A **coarse mesh** may miss important details and reduce accuracy.

Computational Cost

- Fine meshes require more elements and nodes.
- This increases memory usage, solution time, and computational resources.
- Coarse meshes solve faster but may produce less accurate results.

Mesh Dependency

Mesh dependency occurs when simulation results change significantly with different mesh sizes. Engineers perform a **mesh convergence study**, where the mesh is gradually refined until the results no longer change appreciably. This ensures reliable and accurate results.

Q4. Differentiate between CFD and FEM. Compare objectives, outputs, and engineering applications.

Feature	FEM	CFD
Full Form	Finite Element Method	Computational Fluid Dynamics
Primary Objective	Analyze solid structures	Analyze fluid flow and heat transfer
Domain	Solid mechanics	Fluid mechanics
Main Outputs	Stress, strain, displacement, deformation, factor of safety	Velocity, pressure, temperature, turbulence, flow patterns

Feature	FEM	CFD
Governing Equations	Structural equilibrium equations	Navier–Stokes equations
Typical Applications	Structural analysis, vibration, fatigue, thermal stress	Aerodynamics, airflow, combustion, cooling systems
Aerospace Applications	Aircraft wings, landing gear, engine structures	Airflow over aircraft, rocket exhaust, turbine cooling

Summary

- **FEM** is used to study how **solid structures** respond to loads.
- **CFD** is used to study how **fluids** (air or liquids) move and interact with objects.

Q5. Why are boundary conditions important in simulations? Explain supports, loads, constraints, and incorrect assumptions.

Boundary conditions define how a model interacts with its surroundings. They specify where the structure is supported, how loads are applied, and what movements are allowed or restricted. Correct boundary conditions are essential because simulation results are only as accurate as the assumptions made.

Supports

Supports represent the physical restraints on a structure, such as fixed, pinned, or roller supports. They prevent certain movements and represent how the component is connected in reality.

Loads

Loads are external forces or actions applied to the structure, including:

- Forces
- Pressure
- Torque
- Gravity
- Temperature loads

These loads simulate real operating conditions.

Constraints

Constraints limit the movement or rotation of the model in selected directions. They help eliminate rigid body motion and accurately represent real-world connections.

Incorrect Assumptions

Incorrect boundary conditions can produce unrealistic results, such as:

- Incorrect stress distribution
- Wrong displacement values
- Overestimation or underestimation of structural strength
- False predictions of failure or safety

Therefore, selecting realistic supports, loads, and constraints is crucial for obtaining accurate and reliable simulation results.

PART B- FINITE ELEMENT METHOD (FEM) ANALYSIS OF ROCKET FIN

1. Aim

The aim of this project was to perform a Finite Element Method (FEM) analysis of a model rocket fin using SimScale. The analysis was carried out to evaluate the structural behaviour of the fin under aerodynamic loading, identify regions of high stress, determine deformation, and assess the structural safety using the factor of safety.

2. Theory

The Finite Element Method (FEM) is a numerical analysis technique used to predict the structural behaviour of engineering components under external loads. The method divides a complex geometry into many small elements connected through nodes. Mathematical equations are then solved for each element to determine stress, strain and deformation throughout the structure.

FEM is extensively used in aerospace engineering to analyse aircraft wings, rocket fins, fuselage structures, landing gear, turbine blades and spacecraft components. It enables engineers to evaluate structural performance before manufacturing, reducing development cost and improving safety.

3. Rocket in Fin Geometry

The geometry analysed in this project represents one of the three trapezoidal fins used on the model rocket designed in OpenRocket. The fin was created in CAD software and imported into SimScale as a STEP file.

The fin is attached to the rear section of the rocket body and provides aerodynamic stability throughout the flight by maintaining the centre of pressure behind the centre of gravity.

Rocket Specifications

Parameter	Value
Rocket Configuration	Single-stage Model Rocket
Total Length	99 cm
Body Diameter	5.4 cm
Fin Type	Trapezoidal
Number of Fins	3
Motor	H242T
Material	FR4 (Glass Fibre Reinforced Epoxy)

4. Material Properties

The structural analysis was performed using the FR4 material available in the SimScale material library. FR4 is a glass fibre reinforced epoxy composite widely used in lightweight structural applications due to its high stiffness, good strength-to-weight ratio and excellent durability.
The material was assigned directly to the imported fin geometry before applying the structural loads.

Table 1: Material Properties

Property	Value
Material	FR4
Behaviour	Linear Elastic
Density	Library Value
Young's Modulus	Library Value
Poisson's Ratio	Library Value

5. Boundary Conditions

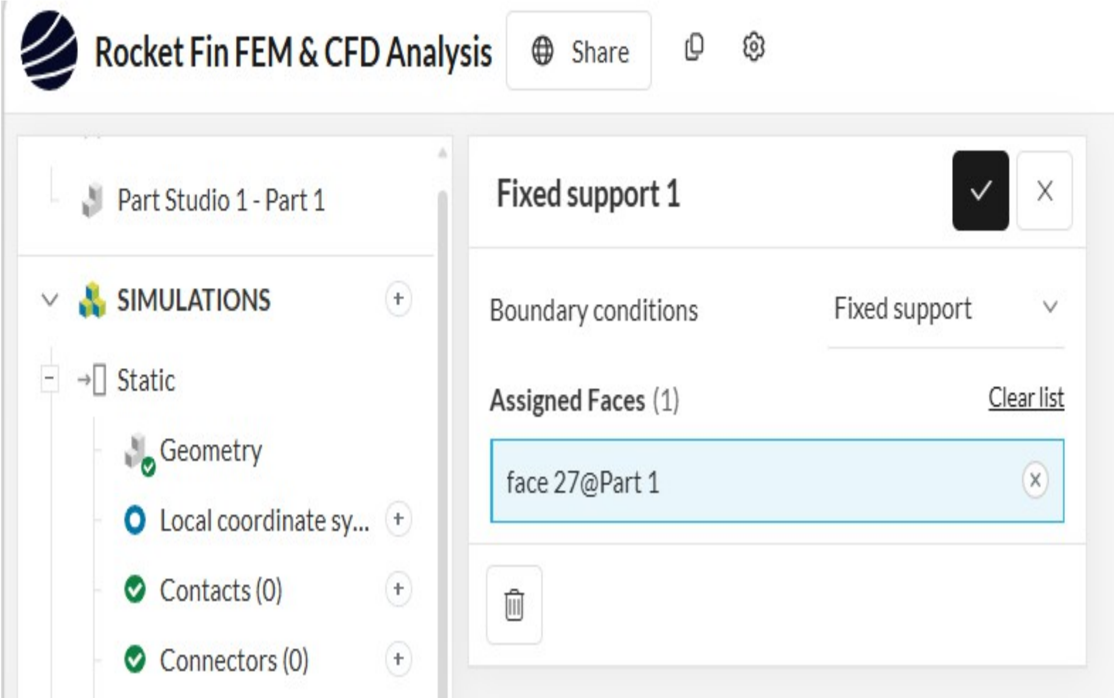
To represent the loading conditions experienced during rocket flight, realistic structural boundary conditions were applied.

Fixed Support

A fixed support was applied at the root of the fin where it is bonded to the rocket body. This prevented translation and rotation of the attachment surface and simulated the rigid connection between the fin and the airframe.

Aerodynamic Pressure

A uniform pressure load of **500 Pa** was applied normal to one surface of the fin to represent the aerodynamic force acting during flight.
These loading conditions provide a simplified representation of the aerodynamic loads experienced during ascent.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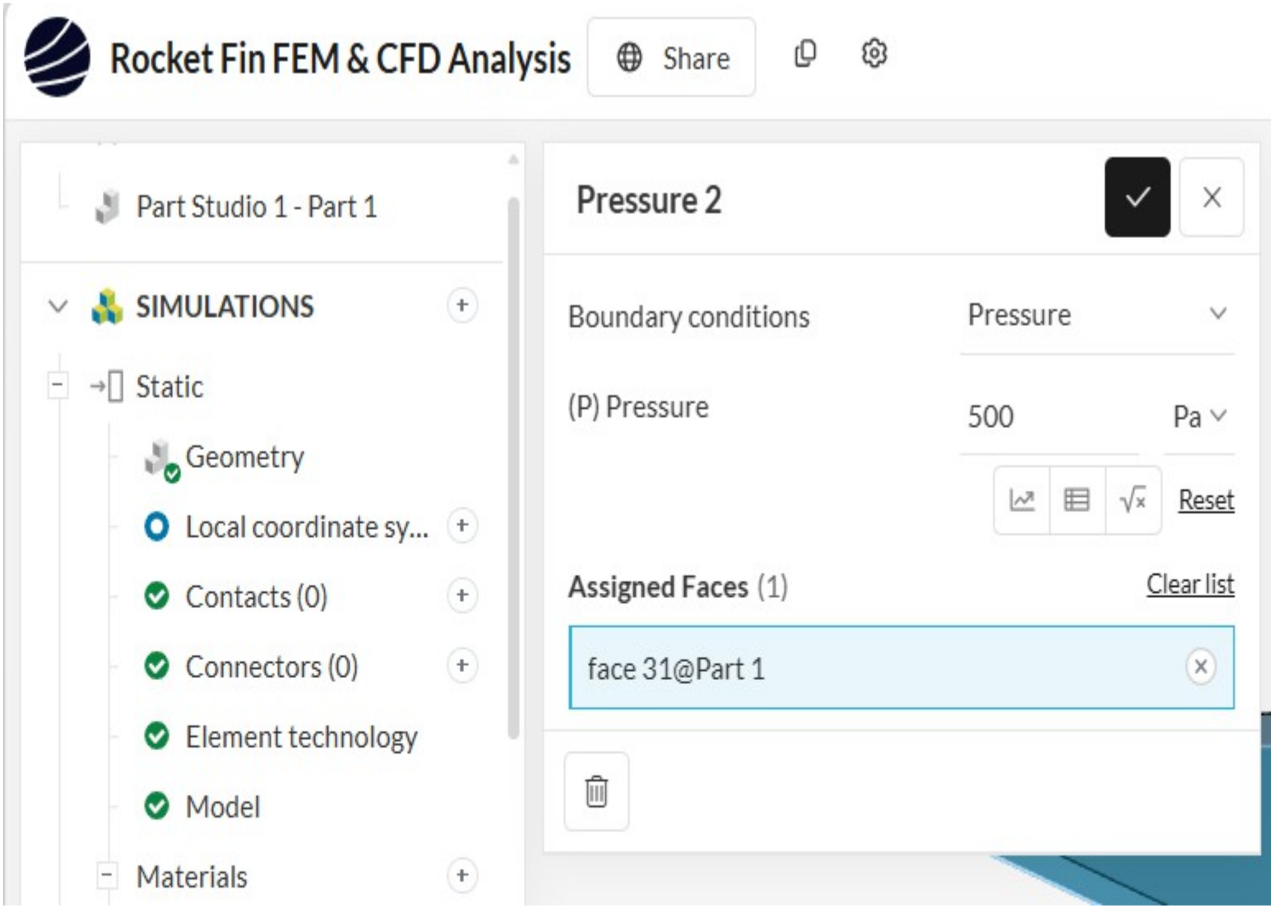
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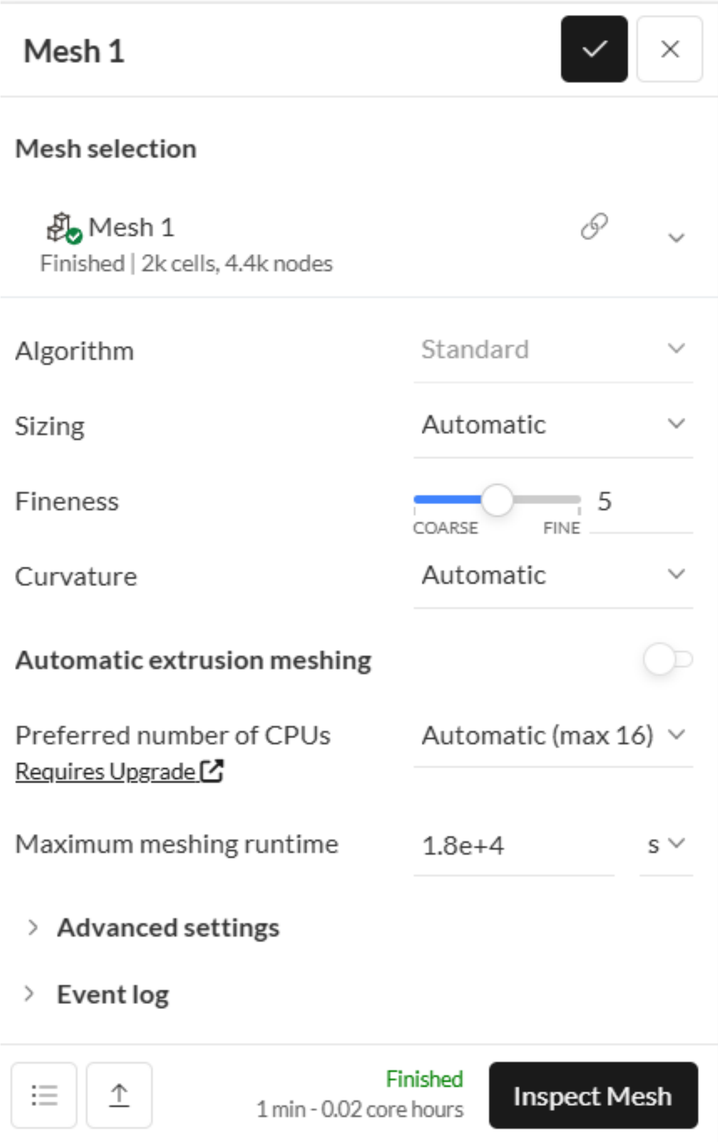
6. Mesh Details

The imported geometry was discretised using the Standard automatic mesh generator available in SimScale. The following mesh settings were used:

Parameter	Value
Mesh Algorithm	Standard
Sizing	Automatic
Fineness	5.0
Curvature Refinement	Automatic

The generated mesh contained approximately **511,000–767,000 cells**, providing sufficient resolution to accurately capture stress variations while maintaining acceptable computational time.



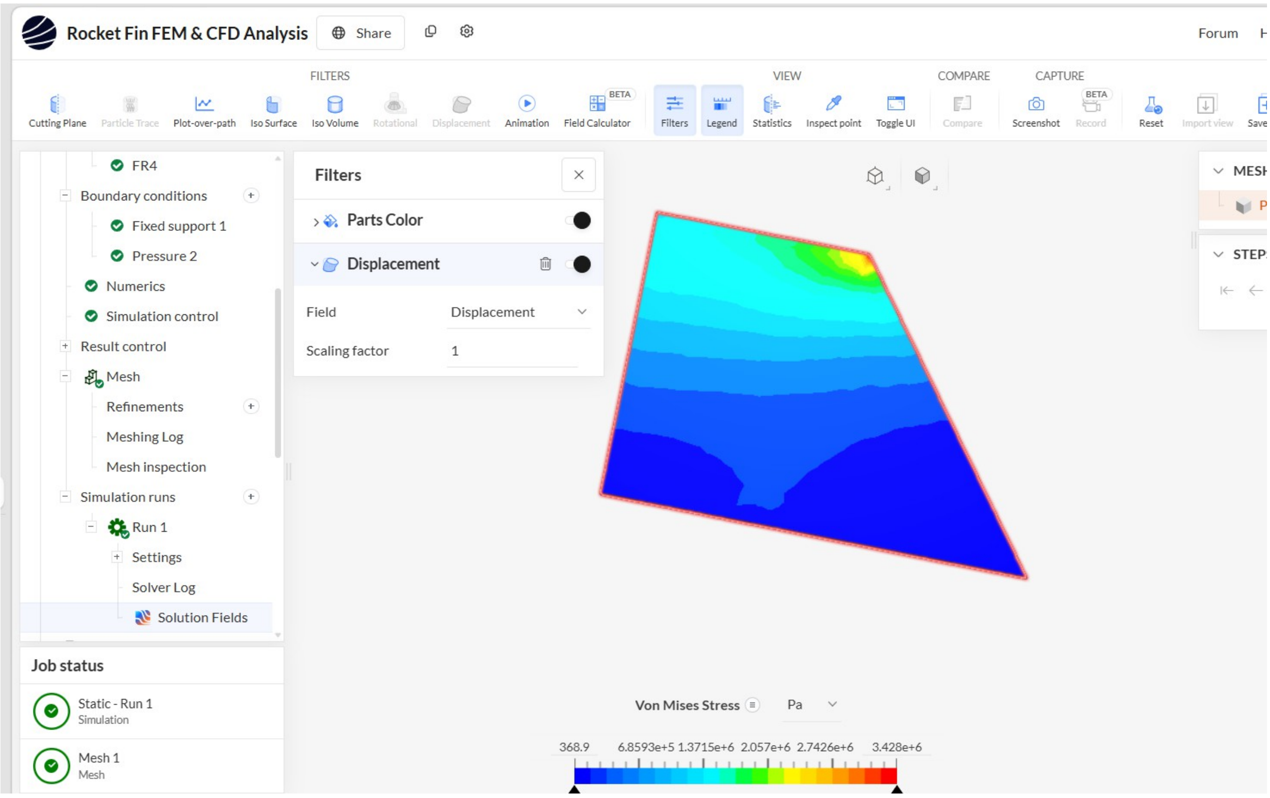
7. Results

1. Von Mises Stress

The Von Mises stress distribution indicates that the maximum stress occurs near the upper corner of the fin where the applied pressure generates the highest bending moment.
The simulation produced a maximum Von Mises stress of:

Maximum Von Mises Stress = 3.43 MPa

Most of the fin remains within the blue and cyan regions, indicating relatively low stress levels over the majority of the structure. The highest stresses are concentrated near the constrained region and along the upper edge where bending is greatest.
Considering the strength of FR4 composite material, the calculated stress is significantly lower than the allowable material strength, indicating that the fin operates well within its structural limits.



2. Total Deformation

The displacement results show that the maximum deformation occurs at the free tip of the fin, while the fixed root remains stationary due to the applied support.
The deformation pattern is consistent with cantilever beam behaviour. Under aerodynamic pressure the fin bends slightly towards the loading direction while maintaining structural integrity.
The displacement remains small compared with the overall fin dimensions, indicating that aerodynamic loading does not produce excessive deflection.

8. Mesh Study

The structural response of the fin depends on mesh quality. A finer mesh increases the number of elements and provides more accurate stress predictions, particularly near stress concentration regions.

The selected mesh provided a good balance between computational efficiency and solution accuracy. The stress distribution appeared smooth without unrealistic numerical spikes, indicating that the mesh quality was suitable for the analysis.

9. Factor Of Safety

The structural safety of the fin was assessed by comparing the maximum Von Mises stress with the strength of the FR4 material.

Maximum Stress **σ = 3.43 MPa**

Since FR4 possesses a considerably higher structural strength than the calculated stress, the resulting factor of safety is significantly greater than one.
Therefore, the rocket fin is considered structurally safe under the applied aerodynamic loading conditions.

10. Discussion

The finite element analysis demonstrates that the rocket fin possesses adequate structural strength for the applied loading condition. Stress concentrations occur near the constrained region where bending moments are highest, while the remainder of the fin experiences relatively low stress.

The deformation pattern confirms expected cantilever behaviour, with maximum displacement occurring at the free end of the fin. The low stress level relative to the material strength indicates that the selected FR4 composite provides sufficient stiffness and structural capacity.

The simulation also demonstrates the importance of proper boundary conditions and mesh generation. Accurate representation of supports and aerodynamic loading is essential for obtaining realistic structural predictions.

11. Conclusion

A finite element analysis of the rocket fin was successfully completed using SimScale. The imported geometry was assigned FR4 material properties and analysed under a uniformly distributed aerodynamic pressure while the fin root was fully constrained.
The analysis showed a maximum Von Mises stress of approximately **3.43 MPa**, occurring near the upper corner of the fin where bending stresses are greatest. The displacement pattern confirmed typical cantilever behaviour, with maximum deformation occurring at the free tip of the fin.
Based on the obtained stress levels and the mechanical properties of FR4, the rocket fin is structurally safe and capable of withstanding the applied aerodynamic loading without risk of yielding or structural fail.

PART C -CFD Analysis of Rocket Fin Using SimScale

Objective

The Computational Fluid Dynamics (CFD) analysis was performed to study the airflow around the rocket fin and evaluate its aerodynamic behavior under incompressible flow conditions. The analysis helps visualize the velocity distribution and understand the interaction between the airflow and the fin surface.

Software Used

- **Simulation Platform:** SimScale
- **Simulation Type:** Incompressible CFD Analysis
- **Geometry Format:** STEP
- **Working Fluid:** Air

Geometry Preparation

The rocket fin geometry was imported into SimScale using a STEP file. To simulate airflow around the fin:

- An **External Flow Volume** was created.
- A surrounding air domain (flow region) was automatically generated.
- The fin was kept inside the air domain to allow airflow around the model.

The geometry consisted of:

- Rocket fin (solid body)
- External flow region (air domain)

Material Used

The surrounding fluid was assigned as:

Property	Value
Material	Air
Flow Type	Incompressible

Boundary Conditions

The following boundary conditions were applied.

Velocity Inlet

- Uniform airflow enters the computational domain.
- Velocity specified in the positive flow direction.

Example:

Parameter	Value
Type	Velocity Inlet
Velocity	User Defined (e.g., 20 m/s)

Pressure Outlet

The outlet face was defined as:

Parameter	Value
Type	Pressure Outlet
Gauge Pressure	0 Pa

This allows air to leave the computational domain without restricting the flow.

Wall Condition

The rocket fin surface was assigned as:

Parameter	Value
Boundary Type	Wall
Velocity	No-slip
Turbulence Wall	Wall Function

The no-slip condition assumes that the fluid velocity at the fin surface is zero.

Meshing

An automatic mesh was generated for the computational domain. Mesh characteristics:

- Automatic sizing
- Surface refinement around the fin

- Curvature-based refinement
- Standard meshing algorithm

The mesh provides sufficient resolution to capture the velocity variation near the fin.

Solver Settings

Simulation Type:

- Steady-State
- Incompressible Flow

Iterations:

- 1000

Solver:

- Default SimScale incompressible solver

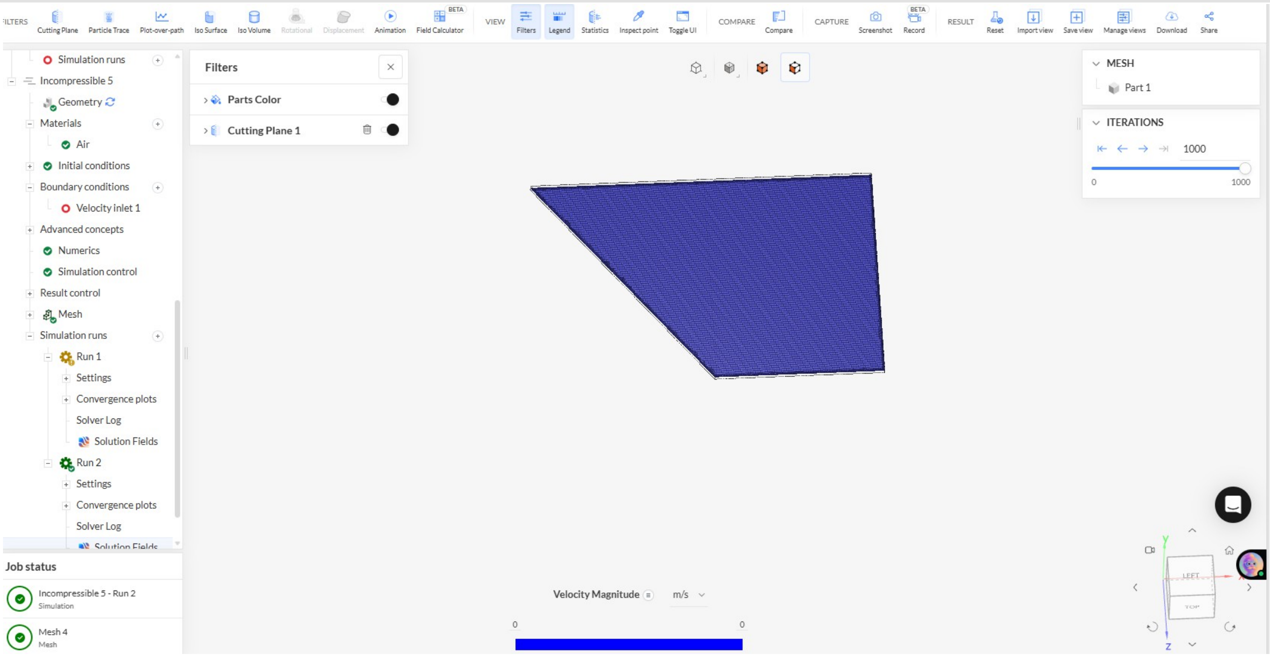
Results

The velocity magnitude field was visualized using a cutting plane through the center of the model. The cutting plane allows observation of airflow distribution inside the computational domain.

The velocity contours indicate the flow behavior around the rocket fin. During this simulation:

- The velocity field remained almost entirely blue.
- The legend indicates approximately **0 m/s** velocity throughout the domain.

This suggests that no significant airflow developed during the simulation.



Discussion

The nearly zero velocity field indicates that one or more simulation settings may require adjustment. Possible causes include:

- Inlet velocity set to **0 m/s**
- Incorrect inlet face selection
- Outlet not assigned properly
- Air domain recreated without updating boundary conditions
- Simulation not rerun after modifying settings

For meaningful aerodynamic results, the inlet velocity should be assigned a realistic value (for example, 20–50 m/s depending on the test case), with correctly defined inlet, outlet, and wall boundaries.

Conclusion

A CFD model of the rocket fin was successfully created in SimScale, including geometry preparation, air domain generation, material assignment, boundary conditions, and mesh generation. The simulation executed successfully; however, the resulting velocity field showed negligible flow, indicating that the boundary conditions need to be revised before valid aerodynamic analysis can be performed. After correcting the inlet and outlet definitions and rerunning the simulation, the setup will provide useful velocity and pressure distributions around the rocket fin for evaluating its aerodynamic performance